

Chapter 24: Eigenvalues and eigenvectors

Definition: Let A be an $n \times n$ matrix. If λ is a scalar and $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^n$ such that

$$A\mathbf{v} = \lambda \cdot \mathbf{v}$$

then we say that λ is an **eigenvalue** of A , and $\mathbf{v}$ is an **eigenvector** of A associated with λ .

Example: Check that $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ are eigenvectors for $A = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix}$, and find the corresponding eigenvalues.

$$A \cdot \mathbf{v}_1 = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A \cdot \mathbf{v}_2 = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 6 \end{bmatrix} = -3 \cdot \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

$\Rightarrow \lambda_1 = 2, \lambda_2 = -3$ are eigenvalues of A .
 $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors of A

Theorem: If $\mathbf{v}$ is an eigenvector of A , then $r\mathbf{v}$ ($r \neq 0$) is also an eigenvector of A .

Proof:

$$\mathbf{v} \text{ is eigenvector of } \Rightarrow A \cdot \mathbf{v} = \lambda \mathbf{v}$$

$$\Rightarrow A(r\mathbf{v}) = rA\mathbf{v} = r(\lambda\mathbf{v}) = \lambda(r\mathbf{v})$$

$$\Rightarrow r\mathbf{v} \text{ is an eigenvector.}$$

Definition: Let λ be an eigenvalue of an $n \times n$ matrix A , and define the **eigenspace** of A corresponding to λ to be

$$E_\lambda(A) = \{\mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \lambda\mathbf{v}\}$$

Theorem: $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$.

Proof: ① $\vec{0} \in E_\lambda(A)$, since $A \cdot \vec{0} = \lambda \cdot \vec{0}$.

② "+" is closed. for $\forall v_1, v_2 \in E_\lambda(A)$,
 $Av_1 = \lambda v_1, Av_2 = \lambda v_2 \Rightarrow A(v_1 + v_2) = \lambda(v_1 + v_2)$

③ By previous thm, scalar mult. is closed.
 $\Rightarrow E_\lambda(A)$ is a subspace.

Finding eigenvalues and eigenvectors

Example: Let $A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$ be a matrix. Find the eigenvalues and eigenvectors of A .

Let $A \cdot v = \lambda v$, where $\vec{v} \neq \vec{0}$.

$$Av = \lambda v = \lambda I \cdot v \Rightarrow (A - \lambda I) \cdot v = \vec{0}.$$

Since $\vec{v} \neq 0$, $\exists$ nonzero sol'n to the hom.

linear system, $\Rightarrow \det(A - \lambda I) = 0$.

$$\Rightarrow \det \begin{bmatrix} 3-\lambda & 0 \\ -1 & -4-\lambda \end{bmatrix} = 0 \Rightarrow (3-\lambda)(-4-\lambda) = 0$$

$\Rightarrow \lambda_1 = 3, \lambda_2 = -4$ are eigenvalues.

$$\textcircled{1} \text{ For } \lambda_1 = 3, (A - 3I) \cdot v = \vec{0}$$

More room for example:

$$\Rightarrow v = s \cdot \begin{bmatrix} -7 \\ 1 \end{bmatrix}, \text{ so } E_3(A) = \text{span} \left\{ \begin{bmatrix} -7 \\ 1 \end{bmatrix} \right\}$$

$$\textcircled{2} \text{ For } \lambda_2 = -4, (A + 4I) \cdot v = \vec{0}$$

$$\Rightarrow v = t \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \text{ so } E_{-4}(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

Definition: Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

be an $n \times n$ matrix. Then

$$\det(\lambda I_n - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}$$

is called the **characteristic polynomial** of A , denoted $c_A(\lambda)$ in your book.

The equation

$$\det(\lambda I_n - A) = 0$$

is called the **characteristic equation** of A .

Theorem: Let A be an $n \times n$ matrix. The eigenvalues of A are the roots of the characteristic polynomial of A .

Example: Find the characteristic polynomial of A if $A = \begin{bmatrix} 2 & 3 & 0 \\ 3 & 2 & 0 \\ -2 & -4 & 4 \end{bmatrix}$, then find the eigenvalues of A .

$$C_A(\lambda) = \det(\lambda I - A)$$

$$= \det \begin{bmatrix} \lambda-2 & -3 & 0 \\ -3 & \lambda-2 & 0 \\ 2 & 4 & \lambda-4 \end{bmatrix}$$

$$= (\lambda-4) \cdot (\lambda-5)(\lambda+1) \Rightarrow \begin{matrix} \lambda_1=4, \lambda_2=5 \\ \lambda_3=-1 \end{matrix}$$

Finding eigenvalues and associated eigenvectors of matrix A

1. Determine the roots of the characteristic polynomial

$$c_A(\lambda) = \det(\lambda I_n - A).$$

These are the *eigenvalues* of A .

2. For each eigenvalue λ , find **all** nontrivial solutions to the homogeneous system

$$(\lambda I_n - A)\mathbf{x} = \mathbf{0}.$$

These are the *eigenvectors* of A associated with λ .

Warning: Do not make the common mistake of first transforming A to RREF and then finding the eigenvalues and eigenvectors of RREF.

Example: Find the eigenvalues and a basis for each eigenspace of $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{bmatrix}$$

$$= (\lambda - 2)^3$$

$\lambda = 2$ is the only eigenvalue.

$$(A - 2I) \cdot \vec{v} = \vec{0} \Rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \vec{0}$$

$$\Rightarrow \vec{v} = s \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow E_2(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$